

Algorithms 2

Linear Programming

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1. Linear Programming

- **Example:**

- In a power grid, we have n power plants and m cities. We want to decide how much electricity each plant should produce to meet the demand at minimum cost.
- Let x_i be the amount of electricity produced by plant i .
- Let b_j be the demand at city j .
- Denote by $a_{i,j}$ the amount of electricity from plant i that is sent to city j .
- For each demand node j , the plants contribute enough power to meet the demand:

$$a_{1,j}x_1 + a_{2,j}x_2 + \dots + a_{n,j}x_n \geq b_j$$

- If each plant also has a cost c_i , then the total cost is $\sum_{i=1}^n c_i x_i$.

- This example suggests the general linear-programming template:

$$\underbrace{\min_{x \in \mathbb{R}^n} c^t x}_{\text{objective}}$$

$$\underbrace{Ax \leq b, \quad x \geq 0}_{\text{constraints}}$$

- So a linear program chooses the values of $x_1, \dots, x_n$ to optimize a linear objective while satisfying linear constraints. More generally, the objective can be written as either a minimization or a maximization problem.

Definition 1.2.1 An Integer program is an optimization problem over $x \in \mathbb{Z}^n$ that can be written in one of the following six forms:

$$\begin{aligned} \max\{c^T x : Ax \leq b\} \\ \min\{c^T x : Ax \geq b\} \end{aligned}$$

$$\begin{aligned} \max\{c^T x : Ax \leq b, x \geq 0\} \\ \min\{c^T x : Ax \geq b, x \geq 0\} \end{aligned}$$

$$\begin{aligned} \max\{c^T x : Ax = b, x \geq 0\} \\ \min\{c^T x : Ax = b, x \geq 0\} \end{aligned}$$

Definition 1.2.2 A linear program is an optimization problem over $x \in \mathbb{R}^n$ that can be written in one of the following six forms:

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 Remark

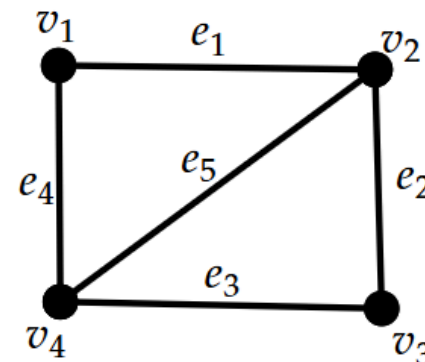
LP's $\in$ P but IP's $\in$ NPH

2. Examples

The *incidence* matrix of a graph G is given by

$$M := M(G) = \begin{matrix} & e_1 & e_2 & \dots & e_m \\ \begin{matrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{matrix} & \begin{bmatrix} & & & \\ & 0/1 & & \\ & & & \\ & & & \end{bmatrix} \end{matrix} \quad M_{ve} := \begin{cases} 1, & v \in e \\ 0, & v \notin e \end{cases}$$

$$\begin{matrix} & e_1 & e_2 & e_3 & e_4 & e_5 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \end{matrix}$$



Each column has $\#1 = 2$

Each row has $\#1 = \deg(v)$

Given G with n vertices and m edges, we seek a vector $\mathbf{x} \in \{0, 1\}^m$ s.t.

- $x_e = 1$ when edge e belongs to the matching, and $x_e = 0$ otherwise.
- maximize $\sum_{e \in E} x_e = \max \mathbf{1}^t \mathbf{x}$.

(We want maximum matching)

This vector $\mathbf{x}$ is the characteristic vector of a matching in G .

So far the above doesn't define a matching!

We need to define constraints.

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We need to define constraints.

- For every vertex $v \in V$: $\sum_{e \in E: v \in e} x_e \leq 1$.

(Every vertex can be matched only once)

2.2 Maximum Matching

Given G with n vertices and m edges, we seek a vector $x \in \{0, 1\}^m$ s.t.

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So far the above doesn't define a matching!

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- For every vertex $v \in V$: $\sum_{e \in E: v \in e} x_e \leq 1$.

(Every vertex can be matched only once)

or by matricial form

$x \in \mathbb{Z}^n$
 $\max \mathbf{1}^t x$

$$\begin{matrix} & e_1 & e_2 & \dots & e_m \\ \begin{matrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & \dots & 0 & 1 \end{bmatrix} \end{matrix}$$

M
 $n \times m$

x
 $m \times 1$

$=$

Mx
 $n \times 1$

$$\begin{bmatrix} (Mx)_1 \\ \vdots \\ (Mx)_i \\ \vdots \end{bmatrix}$$

$(Mx)_i = \langle v_i, x \rangle$
#edges incident to v_i
by the given matching

This is why $Mx \leq 1$
must hold if X is a matching!

For the max-matching problem, we arrive at the following

IP for maximum matching

$$\max_{x \in \mathbb{Z}^m} \{ \mathbf{1}^t x : Mx \leq \mathbf{1}, x \geq \mathbf{0} \}$$

LP for maximum matching

$$\max_{x \in \mathbb{R}^m} \{ \mathbf{1}^t x : Mx \leq \mathbf{1}, x \geq \mathbf{0} \}$$

This called the:
Corresponding LP Relaxation

Remark

We don't need to bound $x \leq \mathbf{1}$ as if there exists i for which $x_i > 1$. Then, there must exist $j \in [m]$ for which $M_{j,i} < 1$. But, we know that M is the *incidence matrix* of G so that for exactly two indices j_1, j_2 we have $M_{j_1,i} = 1$ and $M_{j_2,i} = 1$.

Given G with n vertices and m edges, we seek a vector $\mathbf{y} \in \{0, 1\}^n$ s.t.

- $y_v = 1$ when vertex v belongs to the **vc**, and $y_v = 0$ otherwise.
- minimize $\sum_{v \in V} y_v = \min \mathbf{1}^t \mathbf{y}$.

(We want minimum vertex-cover)

Constraints: $\forall e \in E : \sum_{v \in e} y_v \geq 1$

(Every edge is covered at least once)

This vector $\mathbf{y}$ is the characteristic vector of a vertex cover in G .

2.3 Vertex Cover

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or by matricial form

$\mathbf{y} \in \mathbb{Z}^n$
 $\min \mathbf{1}^t \mathbf{y}$

$$\begin{matrix} & v_1 & v_2 & \dots & v_n \\ \begin{matrix} e_1 \\ e_2 \\ \vdots \\ e_m \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & \dots & 0 & 1 \end{bmatrix} \end{matrix}$$

M^t
 $n \times m$

$\mathbf{y}$
 $n \times 1$

$$= \begin{bmatrix} (M^t \mathbf{y})_1 \\ \vdots \\ (M^t \mathbf{y})_i \\ \vdots \end{bmatrix}$$

$M \mathbf{y}$
 $m \times 1$

$(M^t \mathbf{y})_i = \langle e_i, \mathbf{y} \rangle$
 #vertices covering the edge e_i

This is why $M^t \mathbf{y} \geq \mathbf{1}$
 must hold if $\mathbf{y}$ is a matching!

For the min-vc problem, we arrive at the following

IP for minimum-vc

$$\min_{\mathbf{y} \in \mathbb{Z}^n} \{ \mathbf{1}^t \mathbf{y} : M^t \mathbf{y} \geq \mathbf{1}, \mathbf{y} \geq \mathbf{0} \}$$

LP for minimum-vc

$$\min_{\mathbf{y} \in \mathbb{R}^n} \{ \mathbf{1}^t \mathbf{y} : M \mathbf{y} \geq \mathbf{1}, \mathbf{y} \geq \mathbf{0} \}$$

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What if we wanted vc with minimum weight? i.e. every vertex $v \in V$ has weight $w_v \in \mathbb{R}$?

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$$\min_{\mathbf{y} \in \mathbb{Z}^n} \left\{ \underbrace{\mathbf{w}^t \mathbf{y}}_{\sum_{v \in V} w_v \cdot y_v} : M^t \mathbf{y} \geq \mathbf{1}, \mathbf{y} \geq \mathbf{0} \right\}$$

By Konig we know that in bipartite graphs it must hold

IP for maximum matching

$$\max_{\mathbf{x} \in \mathbb{Z}^n} \{ \mathbf{1}^t \mathbf{x} : M \mathbf{x} \leq \mathbf{1}, \mathbf{x} \geq \mathbf{0} \}$$

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If G is not bipartite, then in fact

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3. Geometric analysis of LP's

Consider the following **LP**:

$$\max x_1 + x_2$$

s.t.

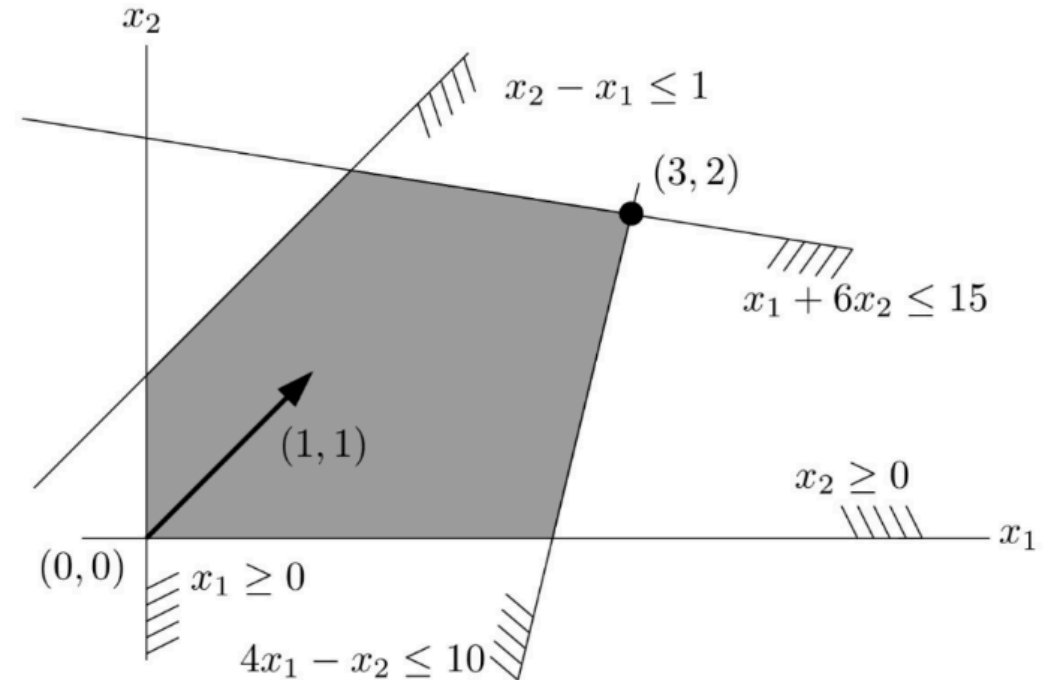
$$-x_1 + x_2 \leq 1$$

$$x_1 + 6x_2 \leq 15$$

$$4x_1 - x_2 \leq 10$$

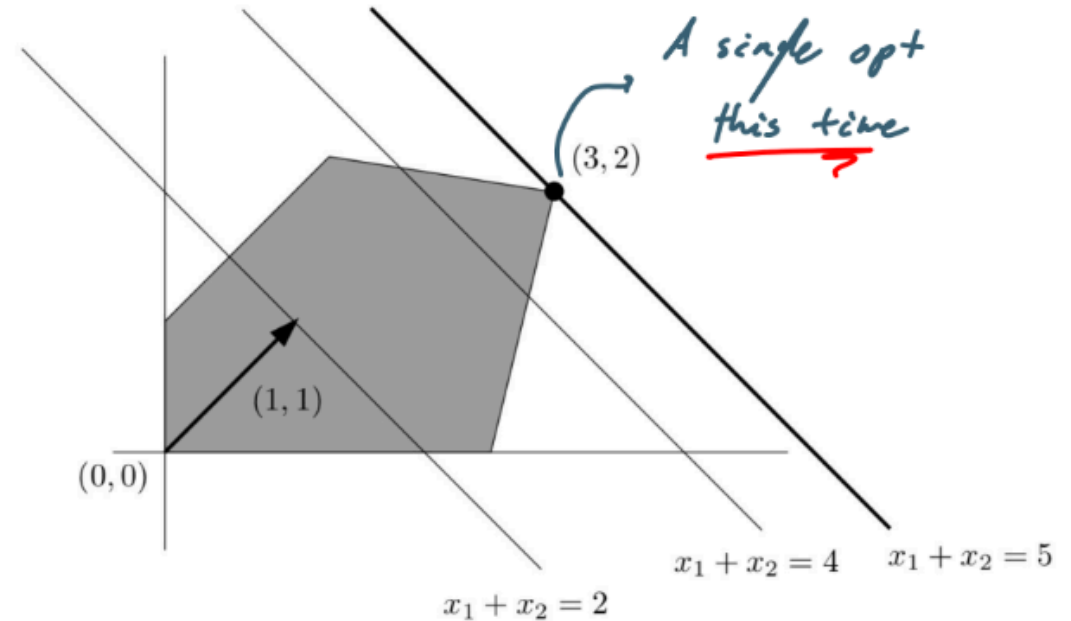
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$$x_2 \geq 0$$

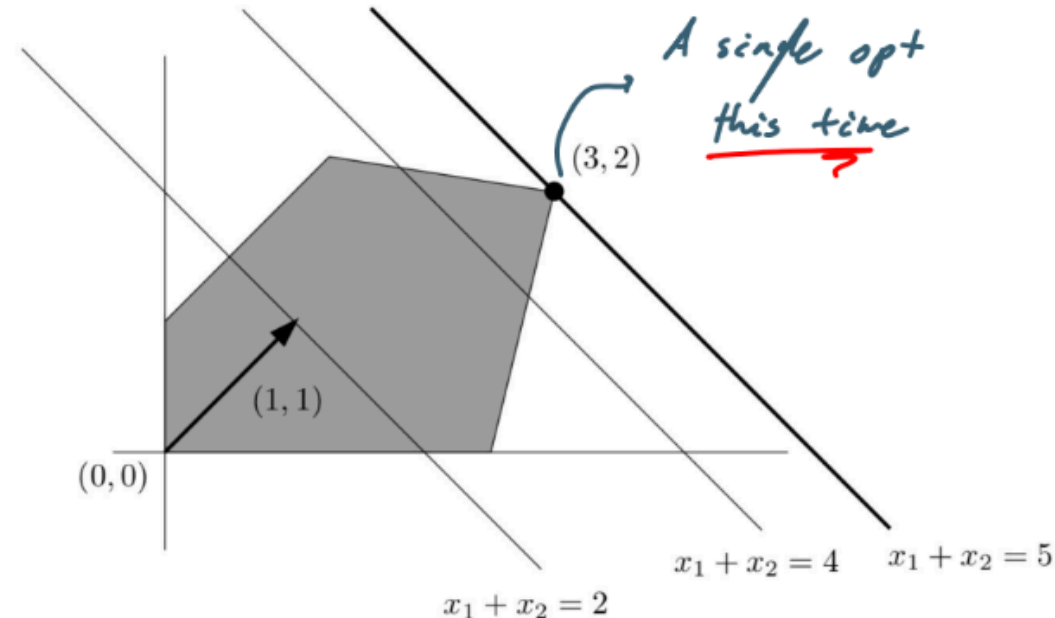
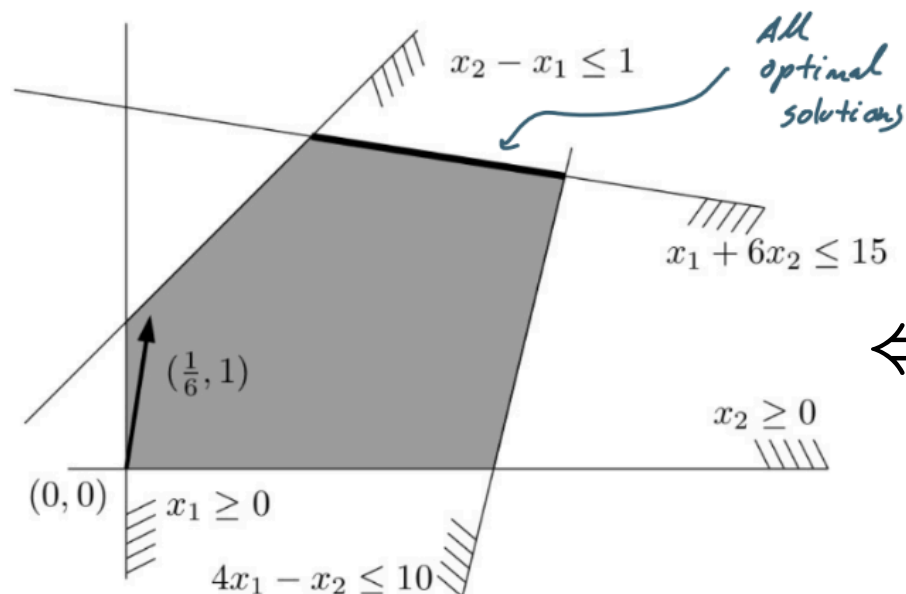


$$\max_{x \in \mathbb{R}^2} (1, 1) \cdot (x_1, x_2) \text{ s.t. } \begin{pmatrix} -1 & 1 \\ 4 & 6 \\ 4 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 1 \\ 15 \\ 10 \end{pmatrix}, \quad (x_1, x_2) \geq (0, 0)$$

Solving the problem Geometrically, would mean taking the line $x_1 + x_2 = c$ and moving it in the direction $(1, 1)$ until we get out of constraint.



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If we change the constraint to $\max \frac{1}{6}x_1 + x_2$.
Then, we have infinitely many solutions.

An LP can fail to have an optimal solution in two ways:

Infeasible — flipping two constraints of our example:

$$\max x_1 + x_2$$

$$x_2 - x_1 \geq 1$$

$$x_1 + 6x_2 \leq 15$$

$$4x_1 - x_2 \geq 10$$

$$x_1, x_2 \geq 0$$

(constraints 1 & 3 flipped)

The constraints have **no common intersection** — the LP is *infeasible*.

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Unbounded — removing two constraints:

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Feasible solutions exist but the objective grows **without bound** — the LP is *unbounded*.

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Theorem 3.2.1

Any **feasible** and **bounded** LP has an **optimal** solution.

Definition 3.3.1 A Linear program in one of the following 2-forms:

$$\max\{c^T x : Ax = b, x \geq 0\}$$

$$\min\{c^T x : Ax = b, x \geq 0\}$$

is said to be in **equational form**;

Remark

Optimal solution for **LP's** in **equational form** are the easiest to explain and we will focus on those.

Any LP can be converted to equational form. Consider a **mixed** LP (not in any standard form):

$$\max 3x_1 - 2x_2$$

$$2x_1 - x_2 \leq 4$$

$$x_1 + 3x_2 \geq 5$$

$$x_1 \geq 0$$

(x_2 is **free** — no sign constraint)

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Final equational LP over $(x_1, y_1, y_2, x_3, x_4 \geq 0)$:

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By solving the final LP we can derive the optimal values for the original LP

3.4 Converting to Equational Form

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Remark

Non-negativity constraints ($x \geq 0$) are called **non-negativity constraints**. The forms with $=$ and $x \geq 0$ are the simplest to analyze.

4. LP's in Equational Form

We study LPs of the form $\max\{\mathbf{c}^T \mathbf{x} : A\mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}\}$ where $A \in M_{m \times n}(\mathbb{R})$ with $\text{rank}(A) = m$ (so $m \leq n$).

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Definition 4.1.1 $\mathbf{x} \in \mathbb{R}^n$ satisfying $A\mathbf{x} = \mathbf{b}$ is called **basic** if $\exists B \subseteq [n]$ with $|B| = m$ s.t.:

- A_B is **non-singular** (columns of A indexed by B are linearly independent), and
- $x_j = 0 \quad \forall j \notin B$.

A basic $\mathbf{x}$ satisfying also $\mathbf{x} \geq \mathbf{0}$ is a **basic feasible solution** (BFS).

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Remark

Basic solutions may **fail** non-negativity – hence the distinction.

$$\underbrace{\begin{bmatrix} -1 & 0 & 1 & 3 & 0 \\ 4 & 0 & 2 & -1 & 1 \\ 0 & 1 & 2 & 0 & -2 \end{bmatrix}}_{A \in \mathbb{R}^{3 \times 5}} \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}}_{\vec{x}} = \underbrace{\begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}}_{\vec{b}} \xRightarrow{B = [1, 2, 4]} \underbrace{\begin{bmatrix} -1 & 0 & 3 \\ 4 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}}_{A_B \in \mathbb{R}^{3 \times 3}} \underbrace{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}}_{\vec{x}_B} = \underbrace{\begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}}_{\vec{b}} \Rightarrow \vec{x} \text{ is basic feasible}$$

Observation 4.1.2 Let $\mathbf{x} \in \mathbb{R}^n$ be feasible. Set $K := \{j \in [n] : x_j > 0\}$. Then

$$\mathbf{x} \text{ is basic} \iff \text{columns of } A_K \text{ are linearly independent.}$$

Proof of observation:

($\implies$): $\mathbf{x}$ is basic, so $\exists B$ with $|B| = m$, A_B non-singular, and $x_j = 0$ for $j \notin B$. Since $\mathbf{x} \geq \mathbf{0}$ we have $K \subseteq B$, so columns of A_K are a subset of the independent columns of A_B – hence independent.

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Proof of observation:

($\implies$): x is basic, so $\exists B$ with $|B| = m$, A_B non-singular, and $x_j = 0$ for $j \notin B$. Since $x \geq 0$ we have $K \subseteq B$, so columns of A_K are a subset of the independent columns of A_B – hence independent.

($\impliedby$): Extend the columns of A_K to a maximal independent set B in A . Since $\text{rk}(A) = m$, we get $|B| = m$, so A_B is non-singular. As $K \subseteq B$, we have $x_j = 0$ for all $j \notin B$. $\square$

Algorithm – to decide if a feasible x is basic:

Input: feasible $x \in \mathbb{R}^n$, i.e. $x \geq 0$ and $Ax = b$.

1. Set $K = \{j \in [n] : x_j > 0\}$.
2. If columns of A_K are independent, return **basic**; else return **not basic**.

Lemma 4.2.1 Any $B \subseteq [n]$ with $|B| = m$ has **at most one** basic feasible solution fitting it.

Proof: Let $N = [n] \setminus B$, and fix a solution $\mathbf{x} \in \mathbb{R}^n$ satisfying $A\mathbf{x} = \mathbf{b}$.

- If $\mathbf{x}$ fits $B \implies$ then $\mathbf{x}_N = \mathbf{0}$.
- so $A_B \mathbf{x}_B + A_N \mathbf{x}_N = \mathbf{b} \implies A_B \mathbf{x}_B = \mathbf{b}$.
- Since A_B is non-singular, $\mathbf{x}_B = A_B^{-1} \mathbf{b}$ is uniquely determined. $\square$

Proposition 4.2.2 There are at most $\binom{n}{m}$ basic feasible solutions.

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This alone is not useful unless optimal solutions are **basic**:

Theorem 4.2.3 Let $\text{LP } \max\{c^T x : x \geq 0, Ax = b\}$ be feasible and bounded.
For every feasible x_0 , there exists a BFS $\bar{x}$ s.t. $c^T \bar{x} \geq c^T x_0$.

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For every feasible x_0 , there exists a BFS $\bar{x}$ s.t. $c^T \bar{x} \geq c^T x_0$.

Applying this to an optimal x_0 (which exists since the LP is feasible and bounded):

$\exists$ an optimal basic feasible solution.

We now have an **exponential algorithm**: enumerate all $\binom{n}{m}$ bases, find each of the **Basic Feasible Solutions**, and pick the best.

5. Geometric View

The linear constraints of a feasible, bounded LP form a **convex polyhedron**:

Convex: any line segment between two feasible points stays inside the region.

Not-convex: a line segment between two feasible points may **exit** the region.

Every half-space $\{x : a^T x \leq b\}$ is convex. The feasible region — as their intersection — is convex too.

The feasible region of an **IP** is generally not convex (integer lattice points only).

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- **Basic feasible solutions** lie on the **vertices** and **edges** of the polyhedron.
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- **Basic feasible solutions** lie on the **vertices** and **edges** of the polyhedron.
- The proposition above guarantees an optimal solution at a **vertex**.

Two classical algorithms traverse the vertices in smarter ways:

Simplex Algorithm

Moves along edges of the polyhedron, improving the objective at each step.
Exponential worst-case, but very fast in practice.

Ellipsoid Algorithm

Approximates the feasible region with shrinking ellipsoids. First provably polynomial-time LP algorithm.

6. Duality

Every LP (the **primal**) has a companion LP called its **dual**.

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We have already seen duality:

König's Theorem: $\tau(G) = \nu(G)$ in bipartite graphs
max matching = min vertex cover

Menger's Theorem: max # of internally disjoint A - B paths = min A - B -cut size

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Why is duality useful?

- It provides **certificates of optimality** for feasible primal solutions.
- It enables LP-based algorithms **without** invoking expensive methods like Simplex or Ellipsoid.
 - Dijkstra's algorithm was designed exactly this way.

Example: Consider the LP

$$\max 5x_1 + 5x_2$$

s.t.

$$3x_1 + x_2 \leq 7$$

$$2x_1 + 4x_2 \leq 8$$

$$x_1, x_2 \geq 0$$

Geometrically, $(1, 2)$ is optimal with objective value $5 \cdot 1 + 5 \cdot 2 = 15$.

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Geometrically, $(1, 2)$ is optimal with objective value $5 \cdot 1 + 5 \cdot 2 = 15$.

Q: How do we **prove** $(1, 2)$ is optimal without drawing a picture?

A: Sum the two constraints (both multiplied by 1):

$$(3x_1 + x_2 \leq 7) + (2x_1 + 4x_2 \leq 8) \Rightarrow 5x_1 + 5x_2 \leq 15.$$

Every feasible solution satisfies $5x_1 + 5x_2 \leq 15$, and $(1, 2)$ achieves 15 – so it is **optimal**. $\square$

Now change the objective to $\max 7x_1 + 4x_2$ (same constraints):

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$$\begin{aligned} & y_1(3x_1 + x_2 \leq 7) + y_2(2x_1 + 4x_2 \leq 8) \\ \Rightarrow & \underbrace{(3y_1 + 2y_2)}_{=7}x_1 + \underbrace{(y_1 + 4y_2)}_{=4}x_2 \leq 7y_1 + 8y_2 \end{aligned}$$

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Solving $3y_1 + 2y_2 = 7$ and $y_1 + 4y_2 = 4$ gives $y_1 = 2, y_2 = \frac{1}{2}$.

Upper bound: $7y_1 + 8y_2 = 7 \cdot 2 + 8 \cdot \frac{1}{2} = 14 + 4 = 18$.

$(2, 1)$ is feasible ($3 \cdot 2 + 1 = 7, 2 \cdot 2 + 4 \cdot 1 = 8$) and achieves $7 \cdot 2 + 4 \cdot 1 = 18$ – so it is **optimal**. $\square$

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2. Multiply constraint i by y_i and **sum**:

$$\sum_i y_i (\mathbf{a}_i^T \mathbf{x}) \leq \sum_i y_i b_i \quad \Rightarrow \quad (A^T \mathbf{y})^T \mathbf{x} \leq \mathbf{y}^T \mathbf{b}.$$

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3. To obtain an upper bound on $\mathbf{c}^T \mathbf{x}$, require $A^T \mathbf{y} = \mathbf{c}$; then $\mathbf{c}^T \mathbf{x} \leq \mathbf{y}^T \mathbf{b}$.

Given the LP $\max\{c^T x : Ax \leq b\}$ (m constraints, n variables):

1. Assign a **dual variable** $y_i \geq 0$ to each constraint $a_i^T x \leq b_i$.
2. Multiply constraint i by y_i and **sum**:

$$\sum_i y_i (a_i^T x) \leq \sum_i y_i b_i \quad \Rightarrow \quad (A^T y)^T x \leq y^T b.$$

3. To obtain an upper bound on $c^T x$, require $A^T y = c$; then $c^T x \leq y^T b$.

Observation 6.2.1 $A \in M_{m \times n}(\mathbb{R})$, $c \in \mathbb{R}^n$, $b \in \mathbb{R}^m$. If $A^T y = c$ and $y \geq 0$, then $c^T x \leq y^T b$ whenever $Ax \leq b$.

Proof: $y^T b \geq y^T (Ax) = (A^T y)^T x = c^T x$. $\square$

6.2 Constructing the Dual

Given the LP $\max\{c^T x : Ax \leq b\}$ (m constraints, n variables):

1. Assign a **dual variable** $y_i \geq 0$ to each constraint $a_i^T x \leq b_i$.
2. Multiply constraint i by y_i and **sum**:

$$\sum_i y_i (a_i^T x) \leq \sum_i y_i b_i \quad \Rightarrow \quad (A^T y)^T x \leq y^T b.$$

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Proof: $y^T b \geq y^T (Ax) = (A^T y)^T x = c^T x$. $\square$

Theorem 6.2.2 Weak Duality Theorem

$$\max\{c^T x : Ax \leq b\} \leq \min\{y^T b : A^T y = c, y \geq 0\}$$

whenever both LPs are feasible and bounded.

The Primal $\rightarrow$ Dual recipe: each primal **constraint** $\rightarrow$ dual **variable**; each primal **variable** $\rightarrow$ dual **constraint**.

Primal constraint $\rightarrow$ dual variable

$$\langle \mathbf{a}_i, \mathbf{x} \rangle \leq b_i \quad \Rightarrow \quad y_i \geq 0$$

$$\langle \mathbf{a}_i, \mathbf{x} \rangle = b_i \quad \Rightarrow \quad y_i \in \mathbb{R}$$

$$\langle \mathbf{a}_i, \mathbf{x} \rangle \geq b_i \quad \Rightarrow \quad y_i \leq 0$$

Primal variable $\rightarrow$ dual constraint

$$x_i \geq 0 \quad \Rightarrow \quad (A^T \mathbf{y})_i \geq c_i$$

$$x_i \leq 0 \quad \Rightarrow \quad (A^T \mathbf{y})_i \leq c_i$$

$$x_i \in \mathbb{R} \quad \Rightarrow \quad (A^T \mathbf{y})_i = c_i$$

Applying the recipe to our LP:

(P)

$$\max 5x_1 + 5x_2$$

$$3x_1 + x_2 \leq 7$$

$$2x_1 + 4x_2 \leq 8$$

$$x_1, x_2 \geq 0$$

$\leftrightarrow$

(D) $\min 7y_1 + 8y_2$

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 Remark

The dual of the dual is the primal.

Weak duality gives an upper bound. The true connection is **much stronger**:

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Theorem 6.4.1 Strong Duality Theorem

Let (P) and (D) be a primal LP and its dual. **Precisely one** of the following holds:

1. Both (P) and (D) are **infeasible**.
2. (P) is **unbounded** and (D) is **infeasible**.
3. (P) is **infeasible** and (D) is **unbounded**.
4. Both (P) and (D) are **feasible** (hence bounded) and their **optimal values coincide**: $\text{OPT}(P) = \text{OPT}(D)$.

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Important remarks:

- Duality does **not** apply to Integer Programming. It applies **only** to LP.

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Case 4 is the main message: when both are feasible, the **duality gap is zero**.

Important remarks:

- Duality does **not** apply to Integer Programming. It applies **only** to LP.
- The LP relaxations of Max Matching and Min VC are **always** dual to one another.
 - This has **no** bearing on their IP versions.
 - In particular, it does **not** imply that $\text{Min VC} \in \text{P}$.

Theorem 6.4.2 Strong Duality Theorem

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Remark

Showing the min-VC LP is the dual of the max-matching LP follows directly from the recipe above.